2.4 Hypothesis and Conclusion

Lesson Introduction

Benchmark	MA.912.LT.4.3 Identify and accurately interpret "if…then," "if and only if," "all" and "not" statements. Find the converse, inverse, and contrapositive of a statement.	Learning Targets	• I can identify the hypothesis of a statement. • I can identify the conclusion of a statement. • I can write the converse, inverse, and contrapositive of a conditional statement.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• What is the relationship between a hypothesis and a conclusion of a statement? • How can I write the converse, inverse, and contrapositive of a conditional statement?
Lesson Narrative	In this lesson, students are introduced to the concept of hypothesis and conclusion within the written form of a conditional statement. Students will engage in an Exploration of what happens when the hypothesis and the conclusion are switched. The new vocabulary "converse," "inverse," and "contrapositive" are introduced with a practice opportunity of writing statements of each.

Component Summary				
	2.4.1 Warm-Up (~5 min.)	Career highlights of a prototype fabricator *(SE p. 98)*	2.4.4 Guided Instruction (15 min.)	Writing converse, inverse, and contrapositive statements *(SE p. 100)*
	2.4.2 Guided Instruction (10 min.)	Writing conditional statements and identifying the hypothesis and conclusion *(SE p. 98)*	2.4.5 Guided Practice (10 min.)	Writing the converse, inverse, and contrapositive for conditional statements *(SE p. 101)*
	2.4.3 Exploration (5 min.)	Exploring what happens when the hypothesis and conclusion are switched *(SE p. 99)*	2.4.6 Wrap-Up (~5 min.)	Cool Down on conditional statements and their converse, inverse, and contrapositive *(SE p. 101)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Conditional statement • Converse statement • Inverse statement • Contrapositive statement <u>Additional Terms:</u> • Hypothesis • Conclusion	N/A	• With these concepts, students often need repetitive practice and examples to help them understand. When reviewing the lesson, consider developing a few additional examples of each that would be relatable to your specific students. • Warm-Up 2.4.1 contains a video to accompany the question prompts.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may write the inverse statement as the converse or vice-versa.
Students may forget to switch the hypothesis, p, and the conclusion, q, when writing a contrapositive statement.
Students may struggle with the notation in the definitions using the hypothesis, p, and the conclusion, q, without a context.
Students may assume that statement written as conditionals are always true.</td><td>This lesson can be very engaging for students. Students often confuse the relationship of the original statement to its converse, inverse, and contrapositive.
Consider asking students if statements and their converses, inverses, and contrapositives are true. Although this is not a requirement of the benchmark it does set up students' understanding of biconditionals.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Co-Craft Questions and Problems
2.4.4 Guided Instruction introduces converse, inverse, and contrapositive. Consider leveraging this routine to help students better understand these concepts. Allow students to create their own converse, inverse, and contrapositive statements at each step of the way.</td><td>MA.K12.MTR.7.1: Real-World Contexts
In 2.4.2 Guided Instruction and 2.4.4 Guided Instruction, students will be engaged in learning about hypotheses and conclusions as well as converse, inverse, and contrapositive statements. Examples of these statements include real-world examples and quotes from famous people to provide for a more relatable entry for students into the concepts.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson is heavy on new vocabulary and introduces students to three concepts: converse, inverse, and contrapositive statements. These can be easy for students to get mixed up. Consider providing a Frayer Model template to allow students to have a reference for future use.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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2.4.1 Warm-Up: Prototype Fabricator

Component Overview: *Play the Career Profile video on the prototype fabricator for the entire class. After students watch the video (3:48 minutes), prompt them to reflect on the profession by answering the two questions independently.*

2.4.1 Warm-Up: Prototype Fabricator

Jeff Tiedeken is a prototype fabricator. He creates metal prototypes through bending, cutting, welding, and assembling.

Jeff states that most people who come in for help do not have an idea of what they need, so his first job is to understand the problem.

1. How does understanding the problem help?
   Answers will vary. Understanding the problem helps him think of ways to solve the problem.

Jeff said that once he understands the problem, he then solves the problem with other people.

2. Why do you think discussing a problem with others can be helpful?
   Answers will vary. Discussing the problem with others allows him to see solutions from other perspectives.

Many students may not know what a prototype fabricator does. Students have many interests and talents yet they may not be aware of various careers. Notice that the follow-up questions purposively do not ask about how Jeff Tiedeken uses mathematics. Students should not be pressured to think that they cannot have a successful career without mathematics. Consider refraining from making this or other career profiles about mathematics. For students who dislike or struggle with mathematics, they may consider these careers as off-limits to them.

2.4.2 Guided Instruction: Hypothesis and Conclusion

Component Overview: *Students will engage in guided instruction on understanding a conditional statement and both a hypothesis and a conclusion. For this activity, consider utilizing a gradual release model through each of the questions.*

2.4.2 Guided Instruction: Hypothesis and Conclusion

Often information is given as an if-then statement. A few examples are shown.

- If you leave the lights on, then it will take longer for people to fall asleep.
- If a goldfish is not exposed to light, then it will lose its color.
- If a quadrilateral has 4 right angles, then it is a rectangle.

1. Write a new if-then statement using a different phrase following "if."
   Answers will vary.

   If *people are on their cell phones at night*, then it will take longer for people to fall asleep.

2. Write a new if-then statement using a different phrase following "then." *Answers will vary.*

   If a quadrilateral has 4 right angles, then *it is not a kite*.

A ***conditional statement*** is a statement in the format of "If P, then Q." The phrase P is the hypothesis. The phrase Q is the conclusion.

3. What is the hypothesis for "If a quadrilateral has 4 right angles, then it is a rectangle"?
   A quadrilateral has 4 right angles.

4. Write a conclusion for each hypothesis.
   Answers will vary.
   a. If yesterday was Sunday, then *today is Monday*.
   b. If you see lightning, then *there is a storm nearby*.

5. William Camden wrote, "The early bird catches the worm." Rewrite Camden's statement as a conditional statement.
   If the bird is early, then it catches the worm.

Discussion questions to consider:

- *Does an if-then statement have to be true?*
- *What is another example of an if-then statement?*
- *Does every if-then statement have to contain both components? Why or why not?*
- *Can the "then" statement exist without there being an "if" statement? Why or why not?*

Understanding the idea of "conditional statement" is an important concept of this lesson. Consider spending extra time reviewing this concept with students to ensure they have a solid understanding, including which part is the hypothesis and which part is the conclusion. Consider adding additional examples that are relatable to your specific students. Consider having students evaluate the truth of a given conditional statement, as all statements written as conditionals are not always true.

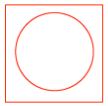
2.4.3 Exploration: Switching the Hypothesis and Conclusion

Component Overview: *Students will explore what happens when the hypothesis and the conclusion of a conditional statement are switched. For this activity, students should complete Questions 1 and 2 on their own and then work with their partner on Question 3.*

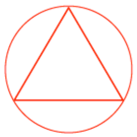
2.4.3 Exploration: Switching the Hypothesis and Conclusion

1. Draw a square or a triangle.

2. If you drew a square, then draw a circle inside the square.



   If you drew a triangle, then draw a circle outside the triangle.



3. Discuss with your partner what happens when the hypothesis and conclusion of each conditional are switched.

   - If you drew a circle inside the square, then draw a square.
   - If you drew a circle outside the triangle, then draw a triangle.

   Summarize your discussion.
   Answers will vary. If you switch the hypothesis, and conclusion, these statements do not make much sense. If they are reworded, then you may be able to still get the same shape. It is not possible to draw a circle inside a square that does not exist yet. It is not possible to draw a circle outside a triangle that does not exist yet.

What students discover during this activity is going to serve as the bridge between what they just learned about conditional statements and what they are about to learn in the next activity with converse, inverse, and contrapositive statements. Do not give away those terms yet in this activity.
Allow a few minutes for students to share out their answers to question 3. This will provide formative data on whether or not students got the intended outcome of this activity. Use this discussion to clear up any misconceptions that may arise.

For students who grasp this quickly and finish early, provide them with a challenge of writing a statement that is true both in the original conditional and when switched.

2.4.4 Guided Instruction: Converse, Inverse, and Contrapositive

Component Overview: *Students are introduced to the vocabulary of converse, inverse, and contrapositive statements. For this activity, engage students in the gradual release model as you guide them through the content of this activity.*

2.4.4 Guided Instruction: Converse, Inverse, and Contrapositive

The ***converse statement*** is a statement in which the given conditional, "If P, then Q," changes to "If Q, then P."

1. Write the converse of "If it is raining, then the grass is wet."
   If the grass is wet, then it is raining.

2. Write a counterexample that shows the converse is not true.
   Answers will vary. A sprinkler may have just been turned off after watering the grass.

The ***inverse statement*** is a statement in which the given conditional, "If P, then Q," changes to "If not P, then not Q."

3. Complete the inverse for "If it is raining, then the grass is wet."
   If it is *not* raining, then the grass is *not* wet.

4. Write a counterexample that shows the inverse is not true.
   Answers will vary. If someone is washing a car on the grass, it may not be raining, however the grass will be wet.

The ***contrapositive statement*** is a statement in which the given conditional, "If P, then Q," changes to "If not Q, then not P."

5. Write the contrapositive for "If it is raining, then the grass is wet."
   If the grass is not wet, then it is not raining.

This section contains several vocabulary terms. Consider the use of a foldable, Frayer Model, anchor chart, or other type of interactive notebook for this section. Whichever method is chosen, it will be important that examples be provided so that students can refer back to them in the future.

While the lesson has specific examples to accompany each statement type, students may need additional examples to connect the content. Prior to releasing students to complete questions 2, 4, and 5, consider providing additional examples that your students can connect to. This can include statements about your town, your school, a famous athlete, a catchy ad slogan, or other relevant examples.

2.4.5 Guided Practice: Converse, Inverse, and Contrapositive

Component Overview: *Students will write the converse, inverse, and contrapositive statements for given conditionals. For this activity, allow students to work independently but check their work with a partner.*

2.4.5 Guided Practice: Converse, Inverse, and Contrapositive

Write the converse, inverse, and contrapositive for each conditional.

1. "If you're not making mistakes, then you're not doing anything." (Quote by John Wooden, an American basketball player and coach)

Converse	*If you're not doing anything, then you're not making mistakes.*
Inverse	*If you are making mistakes, then you are doing something.*
Contrapositive	*If you are doing something, then you are making mistakes.*

2. "If you think you're always in control, then you're not going fast enough." (Quote by Mario Andretti)

Converse	*If you're not going fast enough, then you think you're always in control.*
Inverse	*If you don't think you're always in control, then you are going fast enough.*
Contrapositive	*If you're going fast enough, then you don't think you're always in control.*

Launch this activity with the appropriate level of scaffolding based on student understanding from the prior activity. For example, if most students are struggling with the concepts, consider modeling an example of one of the three in the practice and then releasing. If students generally did well, release them to try them on their own.
Consider implementing a Think-Pair-Share protocol for this activity. Allow students to try the problem on their own and then check their work with a partner.

Use the time for this activity to circulate the room to see how students are completing this table. Try to see as many in question 1 as possible, as any errors students make in that question, they are likely to make in question 2.

2.4.6 Wrap-Up: Cool Down

Component Overview: *Students will complete a Cool Down that assesses their understanding of conditional statement, hypothesis and conclusion, and converse, inverse, and contrapositive statements. This activity should be done independently to provide specific student-level data for additional support.*

2.4.6 Wrap-Up: Cool Down

1. Rewrite the following quote by Michelle Obama as a conditional statement.

   “When someone is cruel or acts like a bully, you don't stoop to their level.”
   If someone is cruel or acts like a bully, then you don’t stoop to their level.

2. Complete the table for the conditional statement shown.

   If you have ridden the roller coaster SheiKra, then you have been to Busch Gardens.

Hypothesis	*You have ridden the roller coaster SheiKra.*
Conclusion	*You have been to Bush Gardens.*

3. Write the converse, inverse, and contrapositive for the conditional statement shown.

   If a fruit is yellow, then it is a banana.

Converse	*If a fruit is a banana, then it is yellow.*
Inverse	*If a fruit is not yellow, then it is not a banana.*
Contrapositive	*If a fruit is not a banana, then it is not yellow.*

This Cool Down leverages each of the three questions to assess a different component of this lesson. Based on student performance, you can determine which areas, if any, need additional support and with which students.

- *Question 1: Conditional Statements*
- *Question 2: Hypothesis and Conclusion*
- *Question 3: Converse, Inverse, and Contrapositive statements*